Demo Ema Eurusd

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Method . . . 3
1. Verdict in one line . . . 3
2. Kill criteria, written down before the run . . . 3
3. Walk-forward results . . . 3
4. Random-direction baseline . . . 3
5. Cost stress . . . 4
6. Integrity audit . . . 4
6.1 Data provenance . . . 4
6.2 Look-ahead . . . 5
6.3 Unit of analysis . . . 5
6.4 Swap units . . . 5
6.5 Simultaneous positions . . . 5
7. Failures . . . 6
8. Reproducibility . . . 6
9. What this means for you . . . 6
Disclaimer . . . 6

Date: 2026-11-05
Auditor: Bina Labs
Verdict: NO EDGE

> edge -0.0240, CI [-0.1147, 0.0671] includes zero - direction is indistinguishable from chance expectancy -0.0616 R/trade after costs 2 of 5 kill criteria triggered.

Method

Walk-forward evaluation on 32895 bars of EURUSDm H1 from the broker's own feed. Parameters are chosen on a fixed grid using only data that existed before each test window; only out-of-sample numbers are reported. Every signal is paired with a coin-flipped control that keeps the entry bar, the stop distance and the target distance identical, so the only thing that changes is direction. Costs are the real ones: per-bar spread from the feed, slippage on every fill, and overnight swap converted from per-lot to per-unit.

1. Verdict in one line

Out-of-sample expectancy -0.0616 R per trade (95% CI [-0.1180, -0.0059]) over 1712 trades. No edge by the criteria below.

We headline the run matching the audited parameter set, not the best of the 2 recorded. The full spread of recorded out-of-sample expectancies is -1.2529 to -0.0616 R per trade.

2. Kill criteria, written down before the run

A system is declared no edge if any of these hold. They are fixed in advance and are not negotiable after seeing the result.

criterion	result
Out-of-sample expectancy 95% CI includes zero	clear — CI [-0.1180, -0.0059] excludes zero
Random-direction baseline not beaten significantly	TRIGGERED — edge -0.0240, CI [-0.1147, 0.0671] includes zero - direction is indistinguishable from chance
Edge survives on one symbol only	not testable in this audit - only one symbol was evaluated. A single-symbol result is a hypothesis until it is run on the rest of the tested set.
Edge disappears under realistic spread + slippage	TRIGGERED — expectancy -0.0616 R/trade after costs
Fewer than 100 out-of-sample trades	clear — 1712 out-of-sample trades

3. Walk-forward results

date	configs tried	OOS trades	expectancy R	95% CI	win rate	PF	cost/risk
2026-11-05	2	1712	-0.0616	[-0.1180, -0.0059]	0.381	0.902	0.051
2026-11-04	2	1265	-1.2529	[-1.3367, -1.1654]	0.223	0.137	0.047

4. Random-direction baseline

The same signals, same entry bars, same stop distance, same target distance, direction mirrored. If the strategy does not beat this, the direction carries no information.

Computed on the full-history run with the a-priori parameter set {'ema_fast': 5, 'ema_slow': 20, 'atr_period': 14, 'sl_atr': 2.0, 'tp_atr': 3.0, 'min_rr': 1.0, 'max_hold': 288, 'slip_points': 2, 'comm': 0.0} (1998 trades), not on the walk-forward out-of-sample trades. The two numbers are deliberately separate: the walk-forward answers "does it generalise", this answers "is the direction real at all".

	expectancy R/trade
strategy	-0.0697
coin-flip control (direction mirrored, same stops/targets)	-0.0456
control sd	1.2153
edge	-0.0240
95% CI of the edge	[-0.1147, 0.0671]
bootstrap resamples where the edge is negative	68.8%

The edge CI contains zero. The direction is noise.

5. Cost stress

The same signals, same stops, same targets, run against a spread twice as wide and one extra pip of slippage on every fill. A large share of "edges" die here, and a customer who cannot reproduce this table has not been given a real cost model.

configuration	expectancy R/trade	change
as configured	-0.0697	—
spread x2, +1 pip slippage	-0.1154	-0.0458

6. Integrity audit

6.1 Data provenance

check	value
bars	33375
duplicate timestamps	0
non-monotonic timestamps	0
gaps larger than one bar	767 (largest 73 bars)
bars from a different timeframe	480 (24x the requested bar size)
invalid OHLC	0
zero-range bars	0
median spread	0.000080
spread is constant	yes

> The spread is identical on every bar in this series. Any filter that claims to trade a spread regime is filtering noise, not a regime. [!IMPORTANT] 480 of the 33375 rows in this file are not H1 bars at all - they are bars spaced 24x apart. MT5 silently returns higher-timeframe bars for the period where the requested timeframe does not exist. This lab trims them before running anything. An EA written by hand usually does not, and a strategy backtested on those rows is being evaluated on a timeframe it does not use.

6.2 Look-ahead

For each signal we shift every bar strictly after the signal bar by a constant offset, keeping OHLC shape intact. If the decision changes - the signal disappears, or flips direction - the strategy read data that did not exist yet.

CLEAN. 2004 signals in the series; the 40 sampled were re-evaluated after shifting every bar up to 12 bars after their signal bar, with no change in existence or direction.

6.3 Unit of analysis

1998 trades came from 582 independent decisions (3.4 trades per decision). Counting trades as observations inflates the sample size and narrows the confidence interval artificially.

estimand	value	95% CI
per-trade, trades independent	-0.0697	[-0.1208, -0.0178]
per-trade, decisions resampled	-0.0697	[-0.1222, -0.0156]
mean per decision (other question)	0.1997	not used

6.4 Swap units

MT5 quotes swap per lot. The cost model consumes it per unit. Getting this wrong does not raise an error; it silently multiplies the carry by the contract size.

	per lot	per unit	contract size
long	-5.6	-0.000056	100000.0
short	0.0	0.000000	100000.0

Carry as a fraction of price per night: 0.0050%

The per-unit figure is plausible for this instrument.

6.5 Simultaneous positions

The mirror held up to 10 position(s) at once across the 1998 closed trades it booked. Every EA shipped from this lab has an InpMaxOpen input (OB_FVG_EA.mq5: 1), and the simulator now supports the same cap. A strategy that fires repeatedly while a position is open is not trading the account the terminal would have run on - it is trading a portfolio.

This run was executed uncapped. If your EA has InpMaxOpen set (they all do), the peak above is the number you need to compare with it.

Run again with InpMaxOpen = 1 and the identical signal set: 996 trades instead of 1998 (1002 could never have been placed), expectancy -0.0838 R per trade instead of -0.0697 R. The confidence interval on 1998 trades is not a confidence interval on the 996 trades an account could actually hold.

Signals generated: 2004. Trades actually closed: 1998. Orders that never filled: 0. Trades rejected because a position was already open: 0. The last two numbers are the difference between a strategy and a strategy that can be executed.

7. Failures

Every parameter set we tried is recorded, including the ones that lost money. This is the part most vendors leave out, and it is the reason a negative verdict is believable.

2 of 2 recorded runs on this strategy/symbol/timeframe had negative out-of-sample expectancy.

date	configs tried	OOS trades	expectancy R	95% CI
2026-11-05	2	1712	-0.0616	[-0.1180, -0.0059]
2026-11-04	2	1265	-1.2529	[-1.3367, -1.1654]

8. Reproducibility

```
python tools/audit_report.py --strategy demo_ema --symbol EURUSDm --tf H1 --sample 40
--cluster-gap 24 --out audits/demo_ema_eurusdm.md
python -m lab.cli integrity --symbol EURUSDm --tf H1 --strategy demo_ema --sample 40
--cluster-gap 24
python -m lab.cli eval --strategy demo_ema --symbol EURUSDm --tf H1
```

Data: data/raw/EURUSDm_H1.csv. Parameters used: {'ema_fast': 5, 'ema_slow': 20, 'atr_period': 14, 'sl_atr': 2.0, 'tp_atr': 3.0, 'min_rr': 1.0, 'max_hold': 288, 'slip_points': 2, 'comm': 0.0}.

9. What this means for you

The difference between this system and a coin flip is not statistically distinguishable. Whatever the vendor's screenshot shows, on your broker's own data, with your costs, the direction this system picks is not better than chance.

Two things follow. First, any profit you have seen from this system is explainable by luck, by the parameters chosen after the fact, or by a backtest that read the future. Second, the money is not in the signal; it is in the cost structure, and that is measurable.

This report is a measurement of a claim. It is not investment advice, it is not a recommendation to trade or stop trading, and it does not tell you what to do with your account.

Disclaimer

This report is a statistical analysis of a software system. It is not investment advice, a recommendation to trade, or a guarantee of future performance. Past performance, in-sample or out-of-sample, does not predict future results. Trading involves risk of loss.